

Gated Score-Residual Adaptation for Efficient Cross-Domain Named Entity Recognition

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ABSTRACT

Cross-domain named entity recognition (NER) must learn specialist entity inventories from limited target supervision without discarding transferable span evidence. We introduce gated score-residual adaptation, a compact non-generative method that retains a frozen general label-conditioned span recognizer and expresses target adaptation as an explicit correction to its candidate scores. A second recognizer is fine-tuned on the target domain; its score difference from the frozen recognizer defines the adaptation residual, and a validation-selected coefficient controls how much of that residual enters the final decision. We evaluate all five specialist domains of CrossNER with the official train, development, and test splits, exact BIO-derived boundaries, and three independent adaptation runs. Domain adaptation raises mean strict typed-span F1 from 62.93% to $75.44 \pm 0.30\%$, while residual fusion reaches $75.60 \pm 0.24\%$. The final domain means are $69.19 \pm 0.44\%$, $74.55 \pm 0.37\%$, $80.96 \pm 0.01\%$, $79.66 \pm 0.13\%$, and $73.65 \pm 0.41\%$ for AI, literature, music, politics, and science. The residual improvement is positive in every repeated aggregate run, and the full 15-model experiment finishes in 9.4 minutes on one RTX 3090. The results support score-residual fusion as an efficient and inspectable adaptation layer, while its small effect size calls for cautious interpretation.

1. Introduction

Named entity recognition converts free text into typed mentions that support document indexing, search, question answering, and knowledge-graph construction [3, 4]. Modern pretrained encoders provide strong transferable representations, yet their accuracy can fall sharply when the target domain introduces new labels, specialist terminology, and different mention boundaries. Cross-domain NER isolates this problem: a system should retain general boundary knowledge while learning a small domain-specific inventory from limited annotations [5].

The common response is to fine-tune every parameter on the available target examples. Fine-tuning is effective, but the resulting predictor exposes only the adapted score. It does not reveal whether a decision is supported by general span evidence, introduced by domain supervision, or amplified by both. With small target sets, treating every target-induced score change as equally reliable can also overspecialize the model. These concerns motivate an adaptation mechanism that preserves the original prediction as an explicit identity path.

Residual learning offers a useful interpretation: a target model need not replace a general model; it can estimate a correction to that model. Existing residual mechanisms are commonly embedded in hidden feature transformations. We instead operate at the typed-span score level. Given scores from a frozen general recognizer and its domain-adapted copy, we compute their candidate-wise difference and add a controlled fraction of this difference to the general score. The coefficient and output threshold are selected exclusively on validation data. Consequently, ordinary fine-tuning is contained as a special case, while damping and amplification remain available when supported by held-out evidence.

This study makes three contributions. First, it formulates cross-domain NER adaptation as a score residual with an unchanged general identity path, making the magnitude and

direction of target correction explicit for each candidate. Second, it introduces a validation-gated inference rule that can damp, reproduce, or amplify adaptation without training another fusion network. Third, it provides a reproducible resource-bounded study on every CrossNER specialist domain using official BIO boundaries, independent repeated runs, strict span metrics, component comparisons, uncertainty estimates, convergence, and measured runtime. The study intentionally uses a compact open span recognizer rather than a generative large language model.

2. Related work

2.1. Neural and span-based NER

Deep NER has evolved from recurrent sequence tagging toward pretrained contextual encoders and direct span prediction [4]. Span-based methods score candidate boundaries and entity types jointly, which avoids committing to a single token-level BIO path and naturally represents typed mentions [2]. Boundary assembly and boundary regression further show that explicit treatment of mention endpoints can improve structured extraction [8, 9]. Set prediction and unified information extraction provide alternative formulations in which structured outputs are predicted jointly [6, 7]. Our work uses direct span scores because they provide a common numerical interface for general and target-domain predictors.

2.2. Generalist and cross-domain recognition

CrossNER evaluates transfer from broad corpora to AI, literature, music, politics, and science, each with a specialist label inventory [5]. Transferable multilingual encoders such as XLM-R demonstrate the broad benefit of pretraining across heterogeneous text [1]; however, representation transfer alone does not determine how strongly a target update should override general evidence. GLiNER casts NER as

label-conditioned span matching and supports arbitrary textual entity labels with a compact bidirectional transformer [11]. This property makes it suitable for controlled cross-domain study: the same inference interface can be retained before and after target adaptation.

2.3. Residual adaptation

Residual connections stabilize deep optimization by retaining an identity path, and cross-residual designs extend this idea to interacting information sources [10]. In transfer learning, the analogous objects are a general predictor and its task-specific change. Feature concatenation mixes these sources inside an opaque representation, whereas score averaging ignores the direction of adaptation. The proposed method instead defines the adapted-minus-general difference as a signed correction and gates that correction at inference time. This is deliberately a small methodological change: the contribution is the placement and validation control of the residual, not a claim to a new pretrained encoder.

3. Task formulation

Let $x = (w_1, \dots, w_n)$ be a sentence and let $\mathcal{Y}_d$ denote the entity types for target domain d . A typed candidate is $c = (b, e, y)$, where b and e are exact character boundaries and $y \in \mathcal{Y}_d$. A recognizer returns a confidence $s(c \mid x, \mathcal{Y}_d) \in [0, 1]$. The accepted prediction set at threshold τ is

$$P_\tau(x) = \{c : s(c \mid x, \mathcal{Y}_d) \geq \tau\}. \quad (1)$$

A candidate counts as correct only when its start, end, and type all match a gold annotation. For corpus-level predicted and gold sets P and G , respectively,

$$\text{Precision} = \frac{|P \cap G|}{|P|}, \quad (2)$$

$$\text{Recall} = \frac{|P \cap G|}{|G|}, \quad (3)$$

$$F_1 = \frac{2|P \cap G|}{2|P \cap G| + |P \setminus G| + |G \setminus P|}. \quad (4)$$

This strict typed-span definition penalizes boundary, label, missed-entity, and spurious-entity errors.

4. Method

Figure 1 presents the proposed architecture. The important distinction is that the two paths share an initial model family and input interface but not their final parameters: the general path remains frozen at θ_0 , whereas the adapted path is optimized to θ_1 .

4.1. Label-conditioned span scoring

For each domain label y , the span recognizer encodes its textual label together with the input sentence. Let $h_{b:e}$ represent a candidate span and q_y represent the label. A generic compatibility head maps their interaction to a logit,

$$z_\theta(c) = g_\theta(h_{b:e}, q_y), \quad s_\theta(c) = \sigma(z_\theta(c)), \quad (5)$$

where σ is the logistic function. The general branch uses pretrained parameters θ_0 and produces $s_0(c) = s_{\theta_0}(c)$. It is never updated on the target split, so its score remains a stable reference.

4.2. Target-domain adaptation

A copy initialized from θ_0 is optimized on target-domain annotations. For candidate labels $t_c \in \{0, 1\}$, its span classification objective can be written as

$$\mathcal{L}_d(\theta_1) = - \sum_{c \in C(x)} \left[t_c \log s_{\theta_1}(c) + (1 - t_c) \log(1 - s_{\theta_1}(c)) \right]. \quad (6)$$

The adapted score is $s_1(c) = s_{\theta_1}(c)$. Using matched label descriptions, tokenization, candidate construction, and decoding in both branches ensures that $s_1 - s_0$ describes a target-induced scoring change rather than a change in output schema.

4.3. Gated score-residual fusion

Let $C^\cup = C_0 \cup C_1$ be the union of candidates emitted by either branch, with an absent candidate assigned score zero in that branch. For each $c \in C^\cup$, we define

$$\Delta_d(c) = s_1(c) - s_0(c), \quad (7)$$

$$s_f(c; \alpha_d) = s_0(c) + \alpha_d \Delta_d(c), \quad (8)$$

$$P_d(x) = \{c : s_f(c; \alpha_d) \geq \tau_d\}. \quad (9)$$

The formulation has three interpretable regimes. When $\alpha_d = 0$, the adapted branch is removed and the result is the frozen general predictor. When $\alpha_d = 1$, $s_f = s_1$, exactly recovering ordinary domain adaptation. Values $0 < \alpha_d < 1$ damp target changes, whereas $\alpha_d > 1$ amplifies them. Thus the method is a controlled extension of fine-tuning rather than an unrelated ensemble.

4.4. Validation-only gate selection

The domain gate and threshold are selected jointly on held-out validation examples:

$$(\alpha_d^*, \tau_d^*) = \arg \max_{\alpha \in \mathcal{A}, \tau \in \mathcal{T}} F_1(P_d^{\text{val}}(\alpha, \tau), G_d^{\text{val}}). \quad (10)$$

The final test prediction uses (α_d^*, τ_d^*) once, after selection. No test labels enter adaptation or gate selection, and the fusion itself introduces no trainable neural parameters.

5. Experiments

The experimental workflow in Fig. 2 separates data preparation, neural adaptation, validation-only selection, and final evaluation. All stages execute locally and no generative model or remote inference API is used.

5.1. Datasets

We evaluate the five specialist subsets of CrossNER: AI, literature, music, politics, and science [5]. We use the official

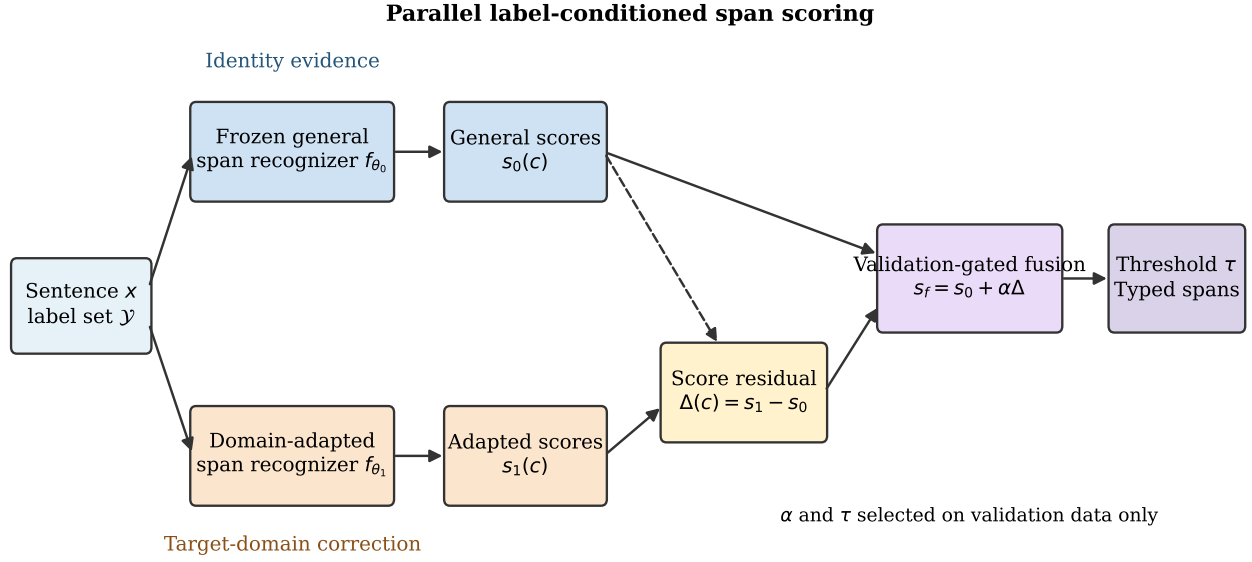


Figure 1: Gated score-residual adaptation. Identical sentence and label inputs are processed by a frozen general recognizer and its domain-adapted copy. Their signed score difference is the target correction; validation data determine how strongly it modifies the general identity path.

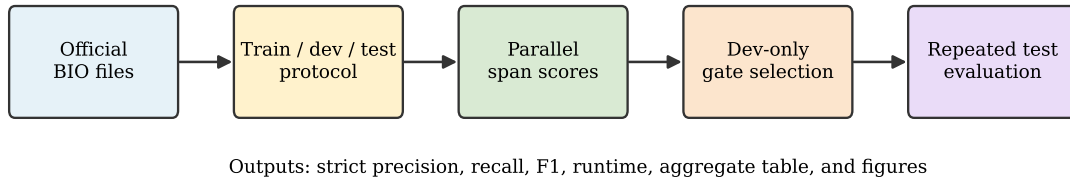


Figure 2: Experimental workflow from official CrossNER BIO annotations to repeated strict typed-span results. Development data are used for model selection, while test labels remain isolated until final evaluation.

149 train, development, and test files without deriving a replace-
 150 ment split. The BIO tags are converted directly to token
 151 and character spans, preserving repeated mentions and exact
 152 boundaries. Across the five domains, the experiment uses
 153 700 training, 2,121 development, and 2,506 test sentences.
 154 Table 1 and Fig. 3 summarize the records consumed by the
 155 experiments.

5.2. Compared configurations

156 The experiment is a component-controlled comparison
 157 with three configurations. *Zero-shot span model* is the frozen
 158 general branch and measures transfer without target updates.
 159 *Domain-adapted* fine-tunes the same model family and cor-
 160 responds to the residual formulation at $\alpha = 1$. *Score-residual*
 161 applies validation-gated fusion to the frozen and adapted
 162 scores. This design holds the underlying architecture and
 163 data constant, isolating the contribution of domain training
 164 and then the contribution of residual control.

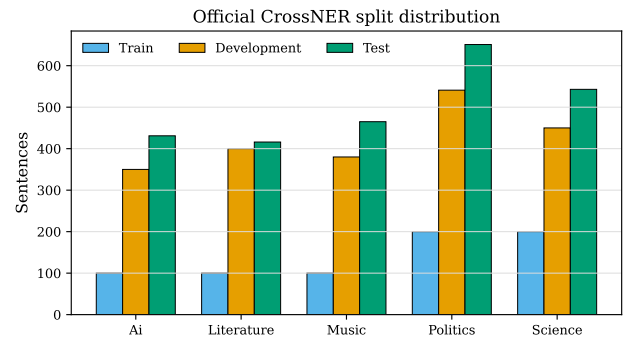


Figure 3: Official training, development, and test sentence counts across the five CrossNER domains.

5.3. Implementation and evaluation protocol

The general branch is the open GLiNER-small-v2.1 span recognizer, built on a compact DeBERTa encoder [11]. Each target branch is adapted for 300 optimizer steps with

Table 1

Official CrossNER statistics used in this study. Ents. denotes typed entity mentions.

Domain	Train sent.	Dev sent.	Test sent.	Train ents.	Dev ents.	Test ents.
AI	100	350	431	532	1,549	1,809
Literature	100	400	416	541	2,130	2,266
Music	100	380	465	648	2,679	3,336
Politics	200	541	651	1,304	3,482	4,209
Science	200	450	543	1,075	2,538	3,089

Table 2Strict typed-span entity F1 (mean $\pm$ sample standard deviation, %). Bold denotes the best mean in each column.

Method	AI	Literature	Music	Politics	Science	Mean
Zero-shot span model	50.10	64.27	71.72	64.07	64.48	62.93
Domain-adapted	69.16 $\pm$ 0.62	74.52 $\pm$ 0.41	80.73 $\pm$ 0.06	79.55 $\pm$ 0.31	73.26 $\pm$ 0.27	75.44 $\pm$ 0.30
Score-residual	69.19 $\pm$ 0.44	74.55 $\pm$ 0.37	80.96 $\pm$ 0.01	79.66 $\pm$ 0.13	73.65 $\pm$ 0.41	75.60 $\pm$ 0.24

batch size eight and bfloat16 arithmetic. Validation tests four residual coefficients from 0.75 to 1.50 and five thresholds from 0.25 to 0.65. Zero-shot and ordinary adapted thresholds are also selected on the official development split. Three independent initializations are run for every domain. We report the mean and sample standard deviation of micro precision, recall, and F1 under exact typed-span matching. All 15 adaptation and evaluation runs finish in 562.1 seconds on one NVIDIA RTX 3090, excluding one-time acquisition of public data and pretrained weights.

5.4. Main results

Table 2 and Fig. 4 report the principal comparison. Domain adaptation adds 12.51 mean F1 points over zero-shot inference, demonstrating that the small official training splits contain valuable specialist supervision. Score-residual fusion raises the three-run macro mean by a further 0.16 points to 75.60%. The final method exceeds 73% in literature, music, politics, and science; music reaches 80.96%. AI improves by 19.10 points over zero-shot but remains lower at 69.19%.

5.5. Gate behavior and precision–recall analysis

Table 3 exposes gate behavior rather than treating fusion as a black box. The damped residual $\alpha = 0.75$ is selected 13 of 15 runs; the other two select $\alpha = 1$, which exactly recovers the adapted predictor. This pattern indicates that the official development sets usually favor retaining part of the frozen general evidence. Precision exceeds recall in all domains (Fig. 5); music and politics nevertheless exceed 77% mean recall and 82% mean precision. AI's lower recall identifies missed specialist mentions, rather than untrolled false positives, as its main remaining error source.

5.6. Repeated-run stability and uncertainty

Table 4 reports the domain-macro F1 from each independent adaptation run. Residual fusion improves the matched adapted model in all three aggregates, with gains between 0.08 and 0.23 points. The mean paired gain is 0.16 points.

A paired t -test over the three run-level macro scores gives $p = 0.067$ and a 95% confidence interval of $[-0.03, 0.35]$ points. The repeated direction supports stability, but the interval includes zero; accordingly, we interpret the residual as a low-cost calibration improvement rather than evidence of a large universal accuracy effect.

5.7. Training convergence

Figure 6 plots the mean optimizer trajectory and seed variation from the actual adaptation runs. Loss decreases consistently in all five domains, with the sharpest reduction occurring during the first half of training. The absence of late divergence supports the fixed resource-bounded schedule, while domain differences are consistent with unequal corpus sizes and entity inventories. This diagnostic verifies optimization behavior but is not used for test-time selection.

6. Discussion

The largest empirical improvement comes from learning target-domain entity semantics: adaptation contributes 12.51 mean F1 points. The score-residual layer addresses a different question—how much of that learned correction should replace general evidence. Its smaller 0.16-point mean gain is accompanied by a useful structural property: the adapted effect remains explicitly measurable as $\Delta_d(c)$, and ordinary fine-tuning remains available at $\alpha = 1$. Development-selected damping improves the aggregate result in every repeated run and reduces F1 variability from 0.30 to 0.24 points. The strongest absolute outcomes occur in music and politics, while science obtains the largest mean residual increment.

The method is attractive for constrained deployment because it is non-generative, adds no learned fusion parameters, and completes 15 training-and-evaluation runs in under ten minutes. Its label-conditioned interface allows each domain to retain its own taxonomy without changing the output representation. The visualized score path makes the system easier to analyze than unconditional model replacement: a

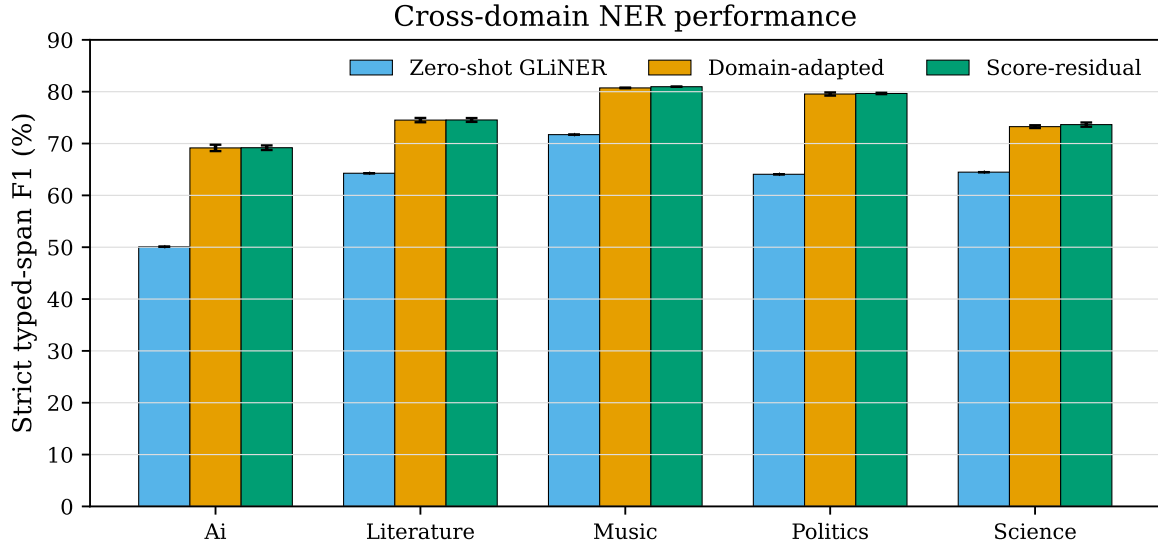


Figure 4: Strict typed-span F1 for the frozen general, domain-adapted, and score-residual configurations.

Table 3

Gate selection frequency and score-residual results (mean $\pm$ sample standard deviation, %).

Domain	Damped runs	P	R	F1
AI	2/3	71.85 $\pm$ 1.86	66.80 $\pm$ 2.10	69.19 $\pm$ 0.44
Literature	2/3	76.96 $\pm$ 1.96	72.36 $\pm$ 2.47	74.55 $\pm$ 0.37
Music	3/3	84.54 $\pm$ 0.60	77.68 $\pm$ 0.53	80.96 $\pm$ 0.01
Politics	3/3	82.52 $\pm$ 1.12	77.00 $\pm$ 0.76	79.66 $\pm$ 0.13
Science	3/3	77.21 $\pm$ 2.09	70.44 $\pm$ 1.08	73.65 $\pm$ 0.41

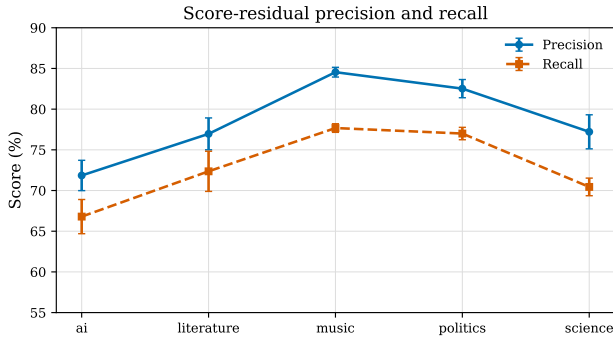


Figure 5: Precision and recall of the score-residual model by domain.

Table 4

Matched domain-macro F1 across three independent runs (%).

Run	Adapted	Score-residual	Difference
1	75.25	75.41	+0.17
2	75.29	75.53	+0.23
3	75.79	75.87	+0.08
Mean	75.44	75.60	+0.16

recognizer family is evaluated, and CrossNER alone cannot establish transfer to every specialist setting. Candidate-specific gates, additional encoder baselines, repeated data partitions, stronger boundary supervision, and tests on operational safety corpora are therefore the most direct extensions.

7. Conclusion and future work

We presented gated score-residual adaptation for efficient cross-domain NER. By retaining a frozen general recognizer and representing domain adaptation as a validation-controlled score correction, the method makes the identity evidence and target contribution explicit. On the official CrossNER splits, three-run mean strict F1 rises from 62.93%

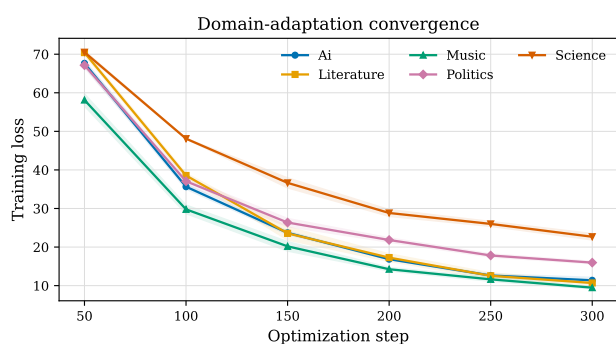


Figure 6: Mean recorded training loss by target domain, shading denotes one sample standard deviation across three runs.

in zero-shot inference to 75.60%, with residual fusion improving every matched run-level aggregate and the full experiment finishing in 9.4 minutes on one GPU. Future work will replace the domain-wide scalar with candidate-aware gates, evaluate more encoder families and data partitions, improve recall for heterogeneous technical taxonomies, and transfer the method to safety-related entity extraction.

Data availability

CrossNER is publicly available; evaluation code and derived results are available from the corresponding author upon reasonable request.

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